



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2022 - 2023)

JEE(Advanced)
REVIEW TEST
30-10-2022

JEE(Main + Advanced) : ENTHUSIAST COURSE [PHASE : S, I(A) & LIVE-I(A)]

Time : 3 Hours

PAPER-2

Maximum Marks : 180

READ THE INSTRUCTIONS CAREFULLY

GENERAL :

1. This sealed booklet is your Question Paper. Do not break the seal till you are told to do so.
2. Use the Optical Response sheet (ORS) provided separately for answering the questions.
3. Blank spaces are provided within this booklet for rough work.
4. Write your name, form number and sign in the space provided on the back cover of this booklet.
5. After breaking the seal of the booklet, verify that the booklet contains **28** pages and that all the **18** questions in each subject and along with the options are legible. If not, contact the invigilator for replacement of the booklet.
6. You are allowed to take away the Question Paper at the end of the examination.

OPTICAL RESPONSE SHEET :

7. The ORS will be collected by the invigilator at the end of the examination.
8. Do not tamper with or mutilate the ORS. **Do not use the ORS for rough work.**
9. Write your name, form number and sign with pen in the space provided for this purpose on the ORS. **Do not write any of these details anywhere else on the ORS.** Darken the appropriate bubble under each digit of your form number.

DARKENING THE BUBBLES ON THE ORS :

10. Use a **BLACK BALL POINT PEN** to darken the bubbles on the ORS.
11. Darken the bubble  **COMPLETELY.**
12. The correct way of darkening a bubble is as : 
13. The ORS is machine-gradable. Ensure that the bubbles are darkened in the correct way.
14. Darken the bubbles **ONLY IF** you are sure of the answer. There is **NO WAY** to erase or "un-darken" a darkened bubble.
15. Take **$g = 10 \text{ m/s}^2$** unless otherwise stated.

QUESTION PAPER FORMAT :

16. The question paper has three parts : Physics, Chemistry and Mathematics.

Please see the last page of this booklet for rest of the instructions

DO NOT BREAK THE SEALS WITHOUT BEING INSTRUCTED TO DO SO BY THE INVIGILATOR

SOME USEFUL CONSTANTS

Atomic No. : H = 1, B = 5, C = 6, N = 7, O = 8, F = 9, Al = 13, P = 15, S = 16, Cl = 17, Br = 35, Xe = 54, Ce = 58

Atomic masses : H = 1, Li = 7, B = 11, C = 12, N = 14, O = 16, F = 19, Na = 23, Mg = 24, Al = 27, P = 31, S = 32, Cl = 35.5, Ca = 40, Fe = 56, Br = 80, I = 127, Xe = 131, Ba = 137, Ce = 140

- | | |
|------------------------------------|--|
| • Boltzmann constant | $k = 1.38 \times 10^{-23} \text{ JK}^{-1}$ |
| • Coulomb's law constant | $\frac{1}{4\pi\epsilon_0} = 9 \times 10^9$ |
| • Universal gravitational constant | $G = 6.67259 \times 10^{-11} \text{ N-m}^2 \text{ kg}^{-2}$ |
| • Speed of light in vacuum | $c = 3 \times 10^8 \text{ ms}^{-1}$ |
| • Stefan-Boltzmann constant | $\sigma = 5.67 \times 10^{-8} \text{ Wm}^{-2}\text{-K}^{-4}$ |
| • Wien's displacement law constant | $b = 2.89 \times 10^{-3} \text{ m-K}$ |
| • Permeability of vacuum | $\mu_0 = 4\pi \times 10^{-7} \text{ NA}^{-2}$ |
| • Permittivity of vacuum | $\epsilon_0 = \frac{1}{\mu_0 c^2}$ |
| • Planck constant | $h = 6.63 \times 10^{-34} \text{ J-s}$ |

Space for Rough Work

HAVE CONTROL → HAVE PATIENCE → HAVE CONFIDENCE ⇒ 100% SUCCESS

BEWARE OF NEGATIVE MARKING

PART-1 : PHYSICS

SECTION-I (i) : (Maximum Marks: 12)

- This section contains **FOUR (04)** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks : +3 If **ONLY** the correct option is chosen.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered)

Negative Marks : -1 In all other cases

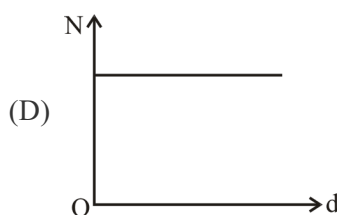
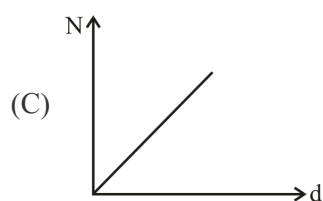
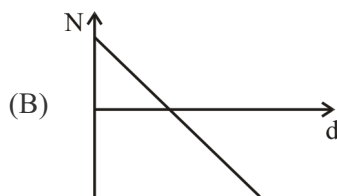
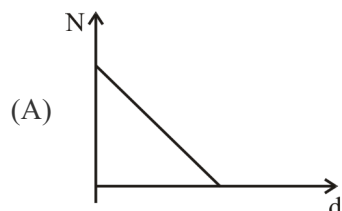
1. In a hydrogen atom like spectrum if an electron de-excite from higher energy level ($n \gg 1$) to a lower energy level then first few frequency of emitted light (Ex : n to $n-1$ or n to $n-2$) are nearly :

(A) In Arithmetic progression

(C) In Harmonic progression

(B) In Geometric progression

(D) In ratio $1^2 : 2^2 : 3^2$
2. A smooth tunnel is made along chord of earth at $\frac{R}{2}$ from centre of earth (R = radius of earth). A object is released from one end of tunnel. Which of the following is correct graph of variation of contact force N vs distance from centre of tunnel :



Space for Rough Work

3. A photon of wavelength λ is incident on photoelectric plate & most energetic electron from plate is now incident on target metal of x ray tube. If cut off wavelength of emitted photons from target material is 2λ then work function of photoelectric plate is :

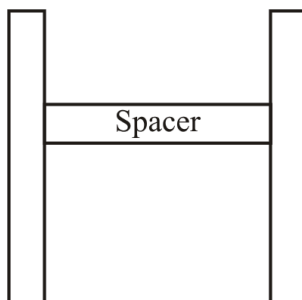
(A) $\frac{hc}{\lambda}$

(B) $\frac{hc}{2\lambda}$

(C) $\frac{hc}{4\lambda}$

(D) $\frac{hc}{8\lambda}$

4. If surrounding temperature is slightly fluctuating as shown in figure. A parallel plate capacity with a spacer is design to operate at constant capacitance. If α_1 & α_2 are coefficient of linear expansion of plate & spacer respectively then. (Neglect area of spacer & it is perfect electric insulator)



(A) $\alpha_1 = \alpha_2$

(B) $\alpha_1 = 2\alpha_2$

(C) $2\alpha_1 = \alpha_2$

(D) $2\alpha_1 = 3\alpha_2$

Space for Rough Work

SECTION-I (ii) : (Maximum Marks: 24)

- This section contains **SIX (06)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.

Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.

Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -2 In all other cases.

- **For Example** : If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

5. If there were only one type of charge in the universe then :

(A) $\oint_S \vec{E} \cdot d\vec{S} \neq 0$ on any surface

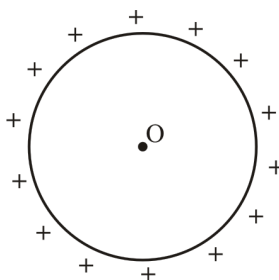
(B) $\oint_S \vec{E} \cdot d\vec{S} = 0$ if charge is outside the surface

(C) $\oint_S \vec{E} \cdot d\vec{S}$ could not be defined

(D) $\oint_S \vec{E} \cdot d\vec{S} = \frac{q}{\epsilon_0}$ if charges of magnitude q were inside the surface.

Space for Rough Work

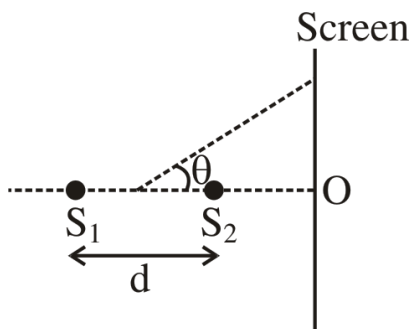
6. A positive charge Q is uniformly distributed along a non conducting circular ring of radius R . A small test charge q of mass m kept on centre of ring. Then :



- (A) If $q > 0$ & is displaced away from center in plane of ring it will be pushed towards center.
- (B) If $q < 0$ & is displaced away from center in plane of ring then it will never return to center.
- (C) If $q < 0$ & is displaced along axis then it will perform SHM.
- (D) If $q < 0$ & is slightly displaced along axis then time period of oscillation is $2\pi\sqrt{\frac{mR^3}{kQq}}$.
7. A real object is kept perpendicular to principal axis of a lens in air. With respect to different cases of image formation by lens, choose **CORRECT** option(s) :
- (A) If magnification is -2.5 then image is real & power of lens is positive.
- (B) If magnification is 0.5 then image is virtual & power of lens is negative.
- (C) If length of image is equal to object then image is real & power of lens is positive.
- (D) If length of image is equal to object then image is virtual & power of lens is negative.
8. A cubical region of space is filled with uniform electric & magnetic fields. A electron & positron enters with same speed v via opposite faces. At this instant :
- (A) The electric forces on both particle causes equal acceleration.
- (B) Acceleration of both particle due to magnetic force are not equal.
- (C) Rate of change of energy for particles is equal.
- (D) Motion of COM (center of mass) is determined by only magnetic force.

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9. In a LCR circuit the average power transferred from a driving source to the circuit is $P = i^2 Z \cos\phi$.
- (A) Here if the power factor $\cos\phi \geq 0$ means $P \geq 0$.
- (B) The driving source gives no energy to oscillator in some cases ($P = 0$)
- (C) The driving source cannot removed or syphon out energy out of oscillator ($P < 0$)
- (D) The driving source can take away energy from oscillator.
10. Two coherent sources of same intensity in same phase S_1 & S_2 of wavelength λ are placed as shown. Distance of screen is much larger than d .



- (A) If $d = \frac{3\lambda}{2}$ then there will be minimum intensity at O.
- (B) If $d = 3\lambda$ then intensity at O is maximum.
- (C) If $d = \frac{11\lambda}{6}$ then intensity at O is 75% of maximum intensity.
- (D) If $d = \lambda$ then minima nearest to O is at $\theta = 60^\circ$.

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SECTION-III : (Maximum Marks: 24)

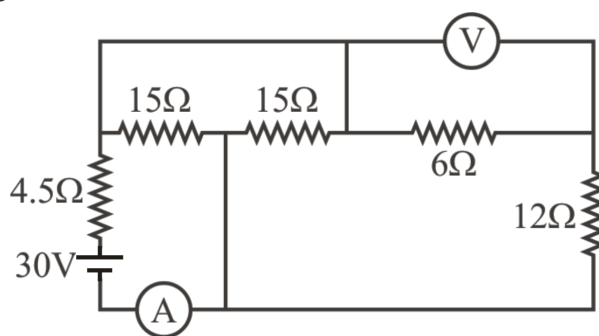
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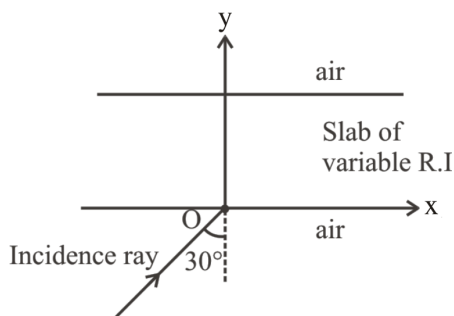
Negative Marks : -1 In all other cases

1. A galvanometer of coil resistance 1Ω is converted into voltmeter by using 5Ω resistance & a similar galvanometer is converted into ammeter using 1Ω resistance. Now ammeter & voltmeter are connected as shown, then ratio of reading of them will be :

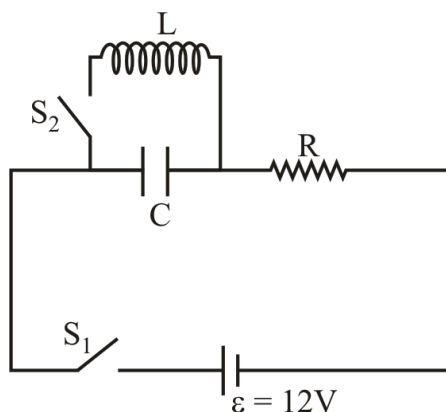


Space for Rough Work

2. A ray of light travelling in air is incident at 30° on one surface of a slab of variable refractive index. If light travels along path $y = 4x^2$ then value of μ at $y = \frac{1}{2}$ inside the slab is $\frac{x}{2}$. Find x .

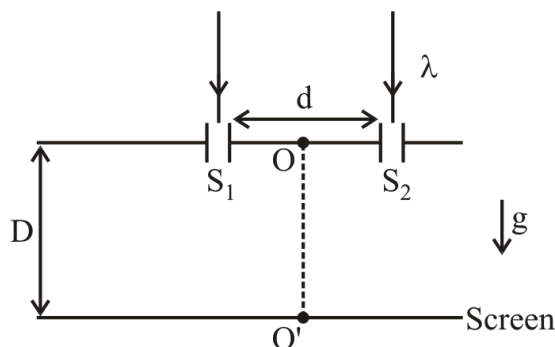


3. A loop made of straight edges has 6 corners at A (0, 0, 0) B (L, 0, 0), C (L, L, 0), D (0, L, 0), E (0, L, L) and F (0, 0, L) is present in a region of magnetic field $\vec{B} = B_0 (\hat{i} + \hat{k})$ T. The flux passing through loop is nB_0L^2 . Find n .
4. Initially at $t = 0$ S_1 is closed & S_2 is open. After time $RC \ln 2$ S_1 is opened & S_2 is closed. At that instant potential difference across capacitor is V_1 . After time $\frac{\pi}{3} \sqrt{LC}$ of closing S_2 potential difference across C is V_2 then $V_1 + V_2$ equal to



Space for Rough Work

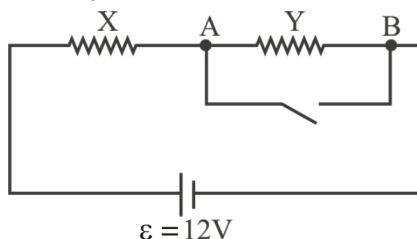
5. If monochromatic light of $\lambda = 5000\text{\AA}$ is incident perpendicular on slits S_1 & S_2 ($d = 1\text{mm}$, $D = 1\text{m}$). If screen is released from rest under gravity then velocity of 2nd maxima w.r.t 1st maxima on screen at $t = 2\text{ sec}$ (in cm/s). (O is taken as origin.)



6. 3 plates A, B & C each of area 1 m^2 are separated by 0.885 mm . Energy stored in system (in μJ) is



7. When a ideal ammeter is put in series in circuit it reads 1 A . When a ideal voltmeter is connected across X it reads 5V . When A & B are shortened by a conducting wire voltmeter reads 8V across cell. Then ratio of resistance X to internal resistance of battery is



8. There exist a time variable magnetic field centred at origin & directed towards negative z-axis $\left(\frac{dB}{dt} = \frac{2\text{T}}{\text{s}}\right)$. A conducting rod is kept in region with coordinates of end point as $(1, 0, 0)\text{ m}$ & $(0, 2, 0)\text{ m}$. Then potential difference between end point of rod is :

Space for Rough Work

PART-2 : CHEMISTRY
SECTION-I (i) : (Maximum Marks: 12)

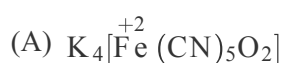
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1. Which of the following is not paramagnetic.

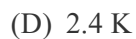
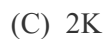
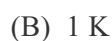
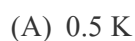
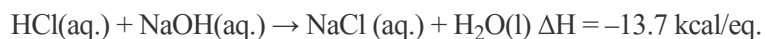


2. In a constant pressure calorimeter 224 ml of 0.1M KOH (aq.) solution is reacted with 50 ml of 0.1M H_2SO_4 (aq.) solution then increase in temperature of solution will be (assume heat capacity of calorimeter is negligible) -

Given :

Specific heat of solution = 1 cal/gm-K

Density of solution = 1 gm/ml



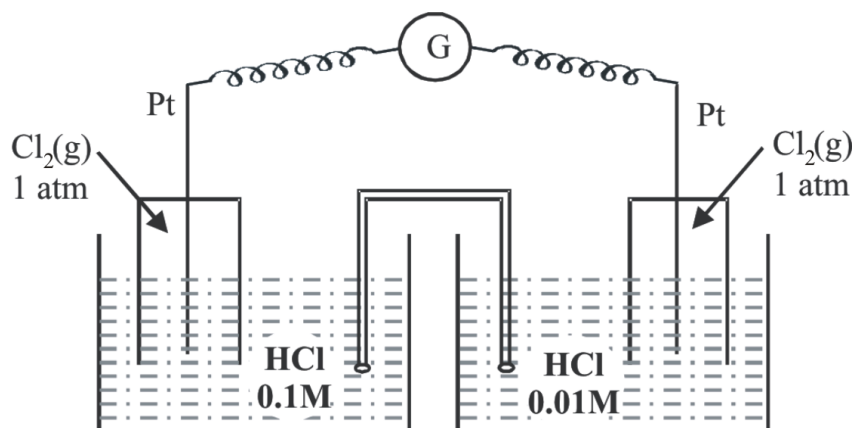
Space for Rough Work

3. For a reaction $C_{(\text{graphite})} + 2H_{2(g)} \longrightarrow CH_{4(g)}$, $\Delta H^\circ = -75 \text{ KJ}$ at 300K and 1 atm. The standard entropies of $C_{(\text{graphite})}$, $H_{2(g)}$ and $CH_{4(g)}$ are 6, 130.6 and 186.2 $\frac{\text{J}}{\text{K mol}}$ respectively. ΔS_{univ} $\left(\text{in } \frac{\text{J}}{\text{K}}\right)$ for the reaction will be :-

(A) -81 (B) 169 (C) -169 (D) 81

4. For the given cell at 25°C, select the correct statement -

Given : $E^\circ_{\text{Cl}_2/\text{Cl}^-} = 1.38 \text{ V}$



- (A) Electrode oxidation potential is negative for anodic compartment.
 (B) Standard oxidation potential is negative for anodic compartment.
 (C) Standard Emf of the cell is zero
 (D) All of these

Space for Rough Work

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5. Which of the following will not give chromyl chloride test.

(A) CCl_4

(B) Hg_2Cl_2

(C) $\text{C}_2\text{H}_5\text{Cl}$

(D) AgCl

6. Which of the following will give Tri basic acid on complete hydrolysis?

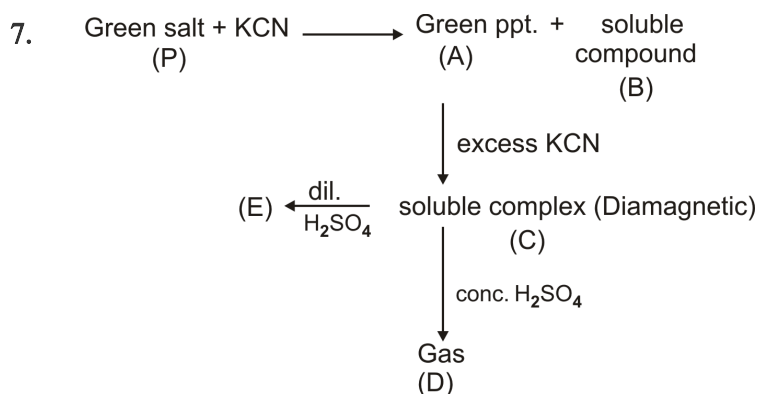
(A) BCl_3

(B) PCl_5

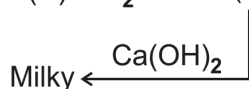
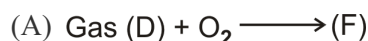
(C) AsCl_3

(D) None

Space for Rough Work



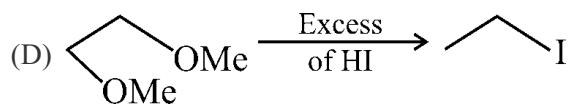
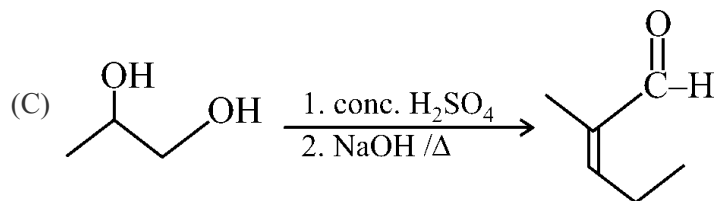
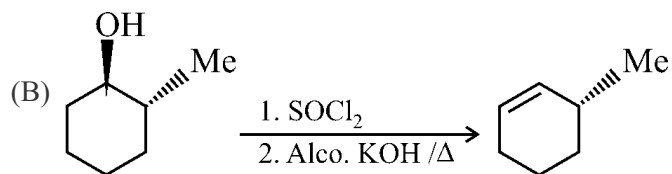
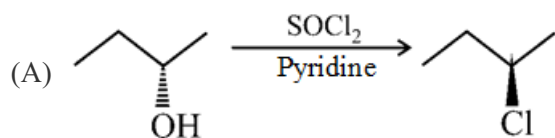
Which option is/are correct for gas (D) and (E).



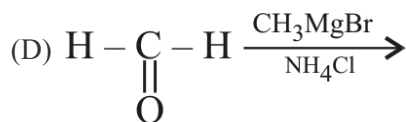
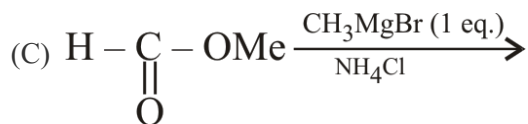
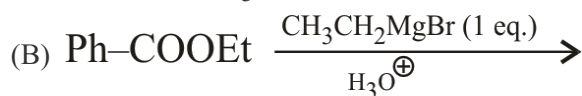
- (B) Gas 'D' has two π bonds
- (C) 'E' has two π bonds & shows H-bonding in its liquid
- (D) Gas 'D' is non poisonous
8. The correct statement(s) related to colloids is(are)
- (A) The process of precipitating colloidal sol by adding an electrolyte is called peptization.
- (B) All colloids may not show Tyndall effect significantly.
- (C) Surfactants form micelle above critical micelle concentration (CMC), if temperature is above Kraft temperature.
- (D) Micelles are macromolecular colloids.

Space for Rough Work

9. Identify reactions correctly match with their products ?



10. Reaction in which product can show positive Haloform test :



Space for Rough Work

SECTION-III : (Maximum Marks: 24)

- This section contains **EIGHT (08)** questions.
- The answer to each question is a **SINGLE DIGIT INTEGER** ranging from 0 to 9, both inclusive.
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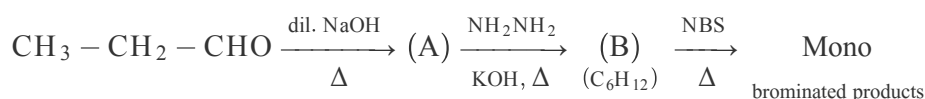
1. Find the number of options which are correct for melamine.
 - (a) Planar (Almost)
 - (b) 15 σ bond
 - (c) 3 π - π bond are present
 - (d) Resonance takes place
 - (e) Pure sp^3 hybridise atoms are = zero
 - (f) All atoms are sp^2 hybridised
 - (g) It is used for making unbreakable Utencils
2. Total number of cation(s) which give white ppt. with $K_4[Fe(CN)_6]$
 Cu^{2+} , Ag^+ , Ca^{2+} , Fe^{3+} , Fe^{2+} , Zn^{2+} , Sr^{2+}
3. The concentration of R in the reaction $R \rightarrow P$ was measured as a function of time and the following data is obtained :

[R] (molar)	1.0	0.75	0.40	0.10
t (min.)	0.0	0.05	0.12	0.18

The order of the reaction is

Space for Rough Work

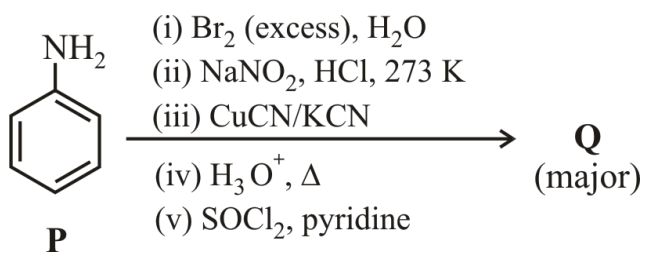
4. Liquids **A** and **B** form ideal solution over the entire range of composition. At temperature T , equimolar binary solution of liquids **A** and **B** has vapour pressure 45 Torr. At the same temperature, a new solution of **A** and **B** having mole fractions x_A and x_B respectively, has vapour pressure of 22.5 Torr. If the value of $\frac{x_A}{x_B}$ in the new solution is Y . Then the value ' $Y-10$ ' is :
- (Given that the vapour pressure of pure liquid A is 20 Torr at temperature T)
5. An alcohol (A) on treatment with conc. H_2SO_4 / Δ gave an alkene (B). The compound (B), on reacting with Br_2/CCl_4 and subsequent didehydrobromination with $NaNH_2(2eq.)$ produced a compound (C). The compound (C) with dil. H_2SO_4 in presence of $HgSO_4$ gave a compound 'D'. The compound D can also be obtained by oxidation of A by pcc or from dry distillation of calcium acetate. If molecular weight of 'C' is $Y \times 10$, then write the value of Y .
6. An organic compound $A(C_{10}H_{18}O_8)$ on treatment with excess of CH_3COCl gives a fully acetylated product whose molar mass is found to be 518 g/mol. How many hydroxyl functional groups are present in A?
7. How many total monobrominated products are formed in following reaction sequence :



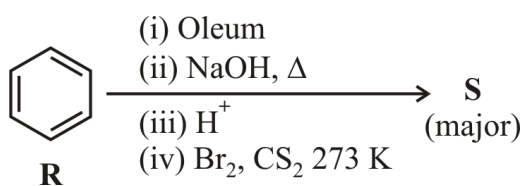
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8. Scheme 1 and 2 describe the conversion of P to Q and R to S, respectively. Scheme 3 describes the synthesis of T from Q and S. The total number of Br atoms in a molecule of T is _____

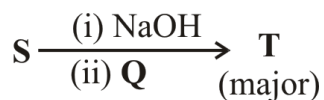
Scheme 1 :



Scheme 2 :



Scheme 3 :



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PART-3 : MATHEMATICS
SECTION-I (i) : (Maximum Marks: 12)

- This section contains **FOUR (04)** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks : +3 If **ONLY** the correct option is chosen.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered)

Negative Marks : -1 In all other cases

1. Let $f : (0, \infty) \rightarrow \mathbb{R}$ be a function such that $f(x) = \lim_{n \rightarrow \infty} \left(1 + \frac{x}{n^2}\right) \left(1 + \frac{2x}{n^2}\right) \left(1 + \frac{3x}{n^2}\right) \dots \left(1 + \frac{nx}{n^2}\right)$.
Then correct option is

(A) $f'(1) > f(1)$	(B) $f(x) < e^x, \forall x \in (0, \infty)$
(C) $f(x) < x^2, \forall x \in (0, \infty)$	(D) $f(c) < \frac{1}{2}$, for some c in $(0, \infty)$
2. Let $f : \mathbb{Z} \rightarrow \mathbb{Z}$ be a function satisfying $f(x) + f(y) = f(x+y) - xy$ and if $f(7) = 21$, then which option is correct - (where $\mathbb{Z}$ is a set of integers)

(A) $f(x)$ is an one-one function
(B) $f(x)$ is an onto function
(C) $f(x)$ is an odd function
(D) Inverse function of $f(x)$ does not exist

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3. $\int \frac{x^3 - x^2 + x - 1}{x^5 + 1} dx = f(x)$ and $f(0) = 0$, then $f(x)$ is

(A) $\frac{1}{5} \ln \left(\frac{1 + x^5}{(1 + x^3)^5} \right)$

(B) $\frac{1}{5} \ln \left(\frac{1 + x^5}{1 + x^3} \right)$

(C) $\frac{1}{5} \ln \left(\frac{1 + x^5}{1 + x} \right)$

(D) $\frac{1}{5} \ln \left(\frac{1 + x^5}{(1 + x)^5} \right)$

4. If $f(x) = \begin{cases} x^n \sin \left(\frac{1}{x^3} \right) & ; x \neq 0 \\ 0 & ; x = 0 \end{cases}$ is continuous but not differentiable at $x = 0$, then correct option is

(A) $n \in (-1, 0)$

(B) $n \in (0, 1]$

(C) $n \in (1, 2]$

(D) $n \in (2, 3]$

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SECTION-I (ii) : (Maximum Marks: 24)

- This section contains **SIX (06)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.

Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.

Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -2 In all other cases.

- **For Example** : If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

5. Let $I(n) = \int_1^n \frac{\sin x}{x^2} dx$ where n is a natural number, then which of the following options are correct -

- (A) $I(2) < \ln 2$ (B) $I(2) > \ln 2 - \frac{1}{4}$ (C) $I(2) > \frac{1}{2\sqrt{2}}$ (D) $I(3) < \ln 3$

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6. Let P_1 and P_2 be two planes given by

$$P_1 : 10x + 15y + 12z - 60 = 0,$$

$$P_2 : -2x + 5y + 4z - 20 = 0.$$

Which of the following straight lines can not be an edge of some tetrahedron whose two faces lie on P_1 and P_2 ?

(A) $\frac{x-1}{0} = \frac{y-1}{0} = \frac{z-1}{5}$

(B) $\frac{x-6}{-5} = \frac{y}{2} = \frac{z}{3}$

(C) $\frac{x}{-2} = \frac{y-4}{5} = \frac{z}{4}$

(D) $\frac{x}{1} = \frac{y-4}{-2} = \frac{z}{3}$

7. If z_1 is a point on locus given by $|z+8| \leq 2$ and z_2 is a point on locus given by $|z+3| + |z-3| \leq 10$ then $|z_1 - z_2|$ can be

(A) 0

(B) 2

(C) 10

(D) 18

8. A and B are two independent events of an experiment, such that $P(A) = \frac{1}{3}$, $P(B) = \frac{1}{4}$, then

(A) $P(A \cup B) = \frac{1}{2}$

(B) $P(A \cup \bar{B}) = \frac{5}{6}$

(C) $P(B \cup \bar{A}) = \frac{3}{4}$

(D) $P(A \cap B) = \frac{1}{12}$

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9. Let $f(x) = \begin{cases} 4x^2 + 1 & x < \frac{1}{2} \\ 2(\log_e x + 1) & \frac{1}{2} \leq x < 1 \\ x^2 + 1 & x \geq 1 \end{cases}$, then which of the following is/are true ?

- (A) $f(x)$ has two points of extremum
- (B) $f(x)$ is continuous $\forall x \in \mathbb{R}$, except at one point
- (C) $f(x)$ is differentiable $\forall x \in \mathbb{R}$ except at one point
- (D) $f(x)$ has exactly one point of minima

10. If area bounded by $y^2 = 8|x|$ and $x^2 = 3|y|$ is Δ , then Δ is divisible is

- (A) 4
- (B) 6
- (C) 8
- (D) 12

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SECTION-III : (Maximum Marks: 24)

- This section contains **EIGHT (08)** questions.
- The answer to each question is a **SINGLE DIGIT INTEGER** ranging from 0 to 9, both inclusive.
- For each question, enter the correct integer value of the answer in the place designated to enter the answer.
- For each question, marks will be awarded in one of the following categories :

Full Marks : +3 If only the correct answer is given.

Zero Marks : 0 If no answer is given.

Negative Marks : -1 In all other cases

1. If $y = f(x)$ satisfy the differential equation $(1 + x^2) f'(x) = x(1 - f(x))$, where $f(0) = \frac{4}{3}$, then value of $f(\sqrt{8}) - \frac{1}{9}$ is equal to
2. Let β be a real number. Consider the matrix $A = \begin{pmatrix} \beta & 0 & 1 \\ 2 & 1 & -2 \\ 3 & 1 & -2 \end{pmatrix}$. If $A^{2022} - (\beta - 1)A^{2021} - \beta A^{2020}$ is a singular matrix, then the value of 9β is _____
3. Let S be the reflection of a point Q with respect to the plane given by $\vec{r} = -(t + p)\hat{i} + t\hat{j} + (1 + p)\hat{k}$ where t, p are real parameters and $\hat{i}, \hat{j}, \hat{k}$ are the unit vectors along the three positive coordinate axes. If the position vectors of Q and S are $10\hat{i} + 15\hat{j} + 20\hat{k}$ and $\alpha\hat{i} + \beta\hat{j} + \gamma\hat{k}$ respectively, then sum of the digits of $(|\alpha| + |\beta| + |\gamma|)$ is

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4. Let z be a complex number with non-zero imaginary part. If $\frac{2 + 3z + 4z^2}{2 - 3z + 4z^2}$ is a real number, then the value of $8|z|^2$ is _____.
5. In a study about a pandemic, data of 900 persons was collected. It was found that
 190 persons had symptom of fever,
 220 persons had symptom of cough,
 220 persons had symptom of breathing problem,
 330 persons had symptom of fever or cough or both,
 350 persons had symptom of cough or breathing problem or both,
 340 persons had symptom of fever or breathing problem or both,
 30 persons had all three symptoms (fever, cough and breathing problem).
 If a person is chosen randomly from these 900 persons and the probability that the person has at least two symptoms is p , then the value of $10p$ is
6. In $\triangle ABC$, let $B(3, 3, -9)$ and $AB = AC$ and altitude AD is $\frac{x-1}{1} = \frac{y-2}{2} = \frac{z+3}{3}$. If co-ordinates of C are (α, β, γ) , then $|\alpha| + |\beta| + |\gamma|$ is
7. Let $y = f(x)$ be a curve such that y intercept of tangent at any point is equal to cube of abscissa of point of tangency. If the curve passes through $\left(1, \frac{1}{2}\right)$, then absolute value of $f(2)$ is
8. The absolute value of $2019 \int_0^{\pi/2} (\cos x)^{2018} \cdot \sin(2020x) dx$

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Space for Rough Work

Space for Rough Work

NAME OF THE CANDIDATE

FORM NO.

I have read all the instructions
and shall abide by them._____
Signature of the CandidateI have verified the identity, name and Form
number of the candidate, and that question
paper and ORS codes are the same._____
Signature of the Invigilator

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